Generated report 2: pathmnist

Objective (as given)

"Balance the rarest tissue class to improve a classifier"

Agent's decision

Task: class_balancing | decision: proceed | task run: class_balancing

profile.json shows 89,996 labelled 28x28x3 images with has_labels true and 9 classes, so class-conditional generation is feasible. The imbalance_ratio is 1.634 (class 6 at 7,886 vs class 8 at 12,885), which is only mild: above the 1.5 cut-off that forces an adapted objective but well below the range where minority-class scarcity usually limits a classifier. With nearly 7,900 real class-6 images, simple random oversampling or class weighting may already be enough, so the evaluation must compare synthetic balancing against real-data resampling, and a null or negative result is a plausible and reportable outcome.

Models run (headline metrics; details on the metrics page)

Run names are <model id>_s<seed>: cvae_s0 is model cvae trained with seed 0. Models in this report: cvae (seed 0); cgan (seeds 0, 1); cdiffusion (seed 0); cdiffusion_long (seed 0). The next page explains seeds, the evaluation space, references, and memorization.

```
cvae_s0          vae       FD 264.6948 (ref 0.2915)  C2ST 0.9998 (ref 0.5081)  label agr. 0.1465
cgan_s0          gan       FD 97.8199 (ref 0.2915)  C2ST 0.9997 (ref 0.5081)  label agr. 0.214
cgan_s1          gan       FD 71.1999 (ref 0.1627)  C2ST 0.9992 (ref 0.502)  label agr. 0.2407
cdiffusion_s0    diffusion FD 42.1907 (ref 0.2915)  C2ST 0.9934 (ref 0.5081)  label agr. 0.8333
cdiffusion_long_s0 not_run_budget
```

Objective test: classifier on held-out real images

```
Target class(es) 6; real-only seed spread (noise level) 0.0063
real_seed0                   target F1 0.9175  macro F1 0.9337
real_seed1                   target F1 0.9112  macro F1 0.9358
real_plus_resampled          target F1 0.9191  macro F1 0.9366
real_plus_cvae_s0            target F1 0.9028  macro F1 0.933
real_plus_cgan_s0            target F1 0.9125  macro F1 0.9359
real_plus_cgan_s1            target F1 0.9107  macro F1 0.932
real_plus_cdiffusion_s0      target F1 0.9072  macro F1 0.9344
```

Downstream-use verdicts

- [not_recommended] Add conditional-diffusion class-6 images to the real training set to raise class-6 F1 (cdiffusion_s0)

- [not_recommended] Add conditional-GAN class-6 images to the real training set to raise class-6 F1 (cgan_s0)

- [not_recommended] Add conditional-VAE class-6 images to the real training set to raise class-6 F1 (cvae_s0)

- [conditional] Balance class 6 by random oversampling of real class-6 training images instead of synthesis (cdiffusion_s0)

Generated report 2: pathmnist (cont.)

Summary (agent-written)

The objective was to balance the rarest PathMNIST tissue class to improve a classifier. The rarest class is class 6 (normal colon mucosa, 7,886 of 89,996 images). The imbalance is mild (1.634x), so the run had to show that synthetic images beat simply oversampling real class-6 images. Three conditional generators (cvae_s0, cgan_s0 and cgan_s1, cdiffusion_s0) were trained on a stratified 20,000-image subsample. Each was used to add 889 class-6 images to the training set. No synthetic arm beat oversampling. On held-out real images, class-6 F1 was 0.9191 with oversampling and between 0.9028 and 0.9125 with synthetic images, against a classifier noise level of 0.0063. Recommendation: do not use these generators for balancing. Random oversampling is the better choice, and with this mild imbalance it is itself within noise of training on real data alone (0.9175 and 0.9112).

How to read this report

Run names and seeds

Run names are <model id>_s<seed>: cvae_s0 is model cvae trained with seed 0. Models in this report: cvae (seed 0); cgan (seeds 0, 1); cdiffusion (seed 0); cdiffusion_long (seed 0).

A seed fixes the model's random initialization, mini-batch order, the validation slice used for early stopping, and the random draws when sampling; the held-out split and preprocessing are the same for every run. Models with two seeds are trained twice as a stability check: a conclusion that holds for both seeds does not depend on one lucky or unlucky initialization. Two runs are far too few to estimate run-to-run variability, so the spread between seeds is not a confidence interval and should not be read as one.

Where realism is measured: the evaluation space

Images are compared in the 128-dimensional feature space of a small CNN classifier trained on the real training images (held-out accuracy 0.94). This plays the role that Inception features play in the usual FID: realism is judged by features that matter for recognizing these images. An ImageNet network would be a poor judge of small, domain-specific images, and too heavy to run here. The classifier never sees synthetic or held-out images during training, and the same classifier scores every model.

Its features only capture what helps tell the classes apart, so two image sets can look alike to it and still differ in ways it ignores. The sample grid figure is the visual check. memorization_pixels repeats the DCR check on raw pixels.

Reference values

Every realism metric is shown with a real-vs-real reference: the same metric computed between a subsample of the training data and the held-out real data. That is the value two genuinely real samples of this size give, so a model is judged by how close it gets to the reference, not to a theoretical ideal.

How memorization is characterized

For each sample, DCR (distance to closest record) is the distance to its nearest training sample in the evaluation space. The held-out real samples give the yardstick: how close a new sample from the same population typically lands to some training sample. dcr_ratio compares the median synthetic DCR to the median held-out DCR (about 1 = as far as new real data; well below 1 = suspiciously close). too_close_fraction counts synthetic samples closer than the 5th percentile of held-out DCRs (about 0.05 is expected without copying).

This detects near-copies of training samples. It is not a privacy guarantee: it does not test membership inference, attribute inference, or any formal property such as differential privacy.

Do generated images belong to the requested class?

label_agreement is the share of synthetic images that the evaluation classifier assigns to the class they were generated for. Its reference is the classifier's accuracy on held-out real images, which is what a perfect class-conditional generator would reach. Low agreement means the model's class conditioning is weak, which matters directly for class balancing.

How to read this report (cont.)

The class-balancing test

The same classifier is trained on several training sets and scored on held-out real images: real data only (twice, with two seeds), real data with the target class oversampled by duplication, and real data with the target class topped up with each model's synthetic images. Synthetic images never enter the held-out set. The difference between the two real-only runs is the noise level; an improvement smaller than that cannot be told apart from training noise. Two runs give a rough yardstick, not a confidence interval.

Objective interpretation

Interpretation

Primary task: class_balancing

Secondary tasks: none

'Balance' and 'the rarest tissue class' map directly to class_balancing; 'to improve a classifier' fixes the success measure as downstream classifier performance on held-out real images rather than sample realism. The rarest class in profile.json is minority_class '6' (7,886 images; normal colon mucosa per DATA_DESCRIPTION.md), so target.class is set explicitly to '6'.

Success criterion

On held-out real PathMNIST images, a classifier trained on real training data plus class-conditional synthetic images of class 6 (normal colon mucosa) reaches a higher class-6 F1 than the same classifier trained on real data with simple random oversampling of class 6, without lowering macro-F1 across the 9 classes.

Whether to model, and how

Decision: proceed (task actually run: class_balancing)

profile.json shows 89,996 labelled 28x28x3 images with has_labels true and 9 classes, so class-conditional generation is feasible. The imbalance_ratio is 1.634 (class 6 at 7,886 vs class 8 at 12,885), which is only mild: above the 1.5 cut-off that forces an adapted objective but well below the range where minority-class scarcity usually limits a classifier. With nearly 7,900 real class-6 images, simple random oversampling or class weighting may already be enough, so the evaluation must compare synthetic balancing against real-data resampling, and a null or negative result is a plausible and reportable outcome.

Other readings considered

- 'Improve a classifier' could mean overall accuracy or macro-F1 rather than minority-class F1; chosen: class-6 F1 as the primary measure with macro-F1 as a guard, because the objective singles out the rarest class.

- 'Balance' could be read as balancing all 9 classes up to the largest (class 8); chosen: only the rarest class, as the objective says 'the rarest tissue class'.

- Because the imbalance is mild (1.634x), the question could be reframed as 'does synthesis beat resampling at all' (proceed_adapted); kept as proceed because the ratio exceeds 1.5, but the resampling comparison is mandatory and the report will treat a no-gain result as the expected baseline outcome.

Data and preprocessing

Dataset (from DATA_DESCRIPTION.md)

PathMNIST, part of the MedMNIST v2 collection. Derived from a histopathology study of colorectal cancer: 28x28 RGB image patches extracted from hematoxylin and eosin (H&E) stained histology slides, downsampled from an original resolution of 224x224. Source: NCT-CRC-HE-100K (train/validation split) and CRC-VAL-HE-7K (test split, from a different clinical center; not loaded by this project). Only the train split is used.

Profile

89,996 samples x 2,352 features; modality image (28, 28, 3)

Sparsity 0.0001; value range [0, 255]; non-negative integers: True

Labels: yes, 9 classes, imbalance ratio 1.634

Held-out split

test_fraction 0.2, stratified: True -> 16000 training / 4000 held-out samples

Reason: After subsampling to 20,000 images, 0.2 leaves 4,000 held-out real images (about 350 of the target class 6 at its 8.8% share in profile.json), enough for per-class F1 and two-sample tests, while keeping 16,000 for training. Stratified because has_labels is true and the downstream test scores class-6 F1, so the held-out set must keep the 9-class mix.

Preprocessing (applied identically for every model; fit on training data only)

- stratified_subsample {"n_samples": 20000}: profile.json has 89,996 images, far more than three conditional image models can train on in a 20-minute budget. With 20,000 images (16,000 train, ~14,400 after the 10% validation hold-back) the timing table gives ~2 s/epoch cVAE, ~8 s/epoch cGAN and ~12.5 s/epoch diffusion, which lets all three fit. Stratified so class 6 keeps its 7,886/89,996 share and the mild 1.634x imbalance is preserved rather than altered by sampling.

- scale_unit_range : Pixels are integer intensities 0-255 (value_range in profile.json, image modality 28x28x3); scaling to [0, 1] matches the Bernoulli-likelihood VAE decoder and the bounded GAN/diffusion outputs, and keeps pixel-space metrics interpretable.

Models, hyperparameters, and training

cvae

Family: vae (conditional); input: transformed; seeds [0]; serves class_balancing; status planned; estimated 2.5 min/seed

Why this model: Fast conditional baseline for generating class-6 (normal colon mucosa) patches, compared against plain random oversampling of real class-6 images because the imbalance is only 1.634x. Revision 2: after the run it is kept only as a negative control. Its samples are blur that ignores the class label (class-6 label agreement 0.0 vs 0.8974 on real class-6 images), so its downstream arm adds mislabeled images and is not evidence about synthetic balancing.

- latent_dim = 32: 32 latent dims for 2,352-pixel 28x28x3 H&E patches with ~14,400 training images: enough to carry stain colour and coarse tissue texture without the latent mostly fitting noise at this sample size.

- base_channels = 32: 32 convolution channels at 28x28 resolution keeps the cVAE the fast baseline (~2 s/epoch on ~14,000 images) so most of the 20-minute budget goes to the GAN and diffusion comparators.

- hidden_dims = [256]: A single 256-unit dense layer between the conv encoder and the 32-dim latent; the 28x28 inputs are small enough that a deeper head would add time without benefit on 16,000 images.

- likelihood = bernoulli: Bernoulli likelihood on H&E pixels scaled to [0, 1] by scale_unit_range. Judgment call: these pixels are continuous intensities, not binary, so this is an approximation; the alternative, a Gaussian decoder, would not keep outputs inside the bounded 0-255 range the profile reports. Revision 2 note: the run produced blurry class-agnostic samples (class-6 label agreement 0.0).

- epochs = 60: Maximum of 60 epochs (~2 min at ~2 s/epoch on ~14,400 images); early stopping on the 10% validation split usually ends sooner, so this is a cap not a target.

- kl_warmup_epochs = 6: 6 epochs, about 10% of the 60-epoch maximum, to ramp the KL weight and avoid posterior collapse while the decoder learns the dominant stain colours.

- batch_size = 128: 128 gives ~112 optimizer steps per epoch on ~14,400 training images, a good balance of gradient noise and MPS throughput for 28x28x3 inputs.

- lr = 0.001: Adam at 1e-3 so the cVAE converges inside its ~2 minute share of the 20-minute budget on ~14,400 PathMNIST training images; the revision-1 validation loss plateaued by about epoch 40 of 60 without oscillating, so the rate was not the limiting factor.

Training (cvae_s0): 60 of 60 max epochs, kept epoch 57 (max epochs reached); 1.745 min on mps; 14400 fit / 1600 validation samples; 1,707,267 parameters

cgan

Family: gan (conditional); input: transformed; seeds [0, 1]; serves class_balancing; status planned; estimated 3.5 min/seed

Why this model: Conditional GAN can produce sharper class-6 patches than the VAE, which may matter for a classifier distinguishing normal mucosa from adenocarcinoma epithelium (class

Models, hyperparameters, and training (cont.)

8). Two seeds because GAN training is unstable and a single run could not separate a real effect on class-6 F1 from a lucky or collapsed run. Revision 2: both seeds failed to condition (discriminator hinge loss rose to about 1.98, overall label agreement 0.214 and 0.241 vs 0.94, class-6 agreement under 0.1), so its downstream arms test label noise rather than synthetic balancing and are reported that way.

- latent_dim = 64: 64-dim noise for the generator, giving enough variation to cover stain differences between slides (DATA_DESCRIPTION.md caveat) within one tissue class.

- base_channels = 64: 64 channels for generator and projection discriminator; GANs need a stronger discriminator than the VAE encoder to produce sharp H&E texture at 28x28.

- epochs = 25: Fixed 25 epochs (~3.3 min per seed at ~8 s/epoch on ~14,400 images); GAN losses do not track sample quality, so there is no early stopping and the count is set by the budget.

- batch_size = 128: 128 images per step, ~112 steps per epoch on ~14,400 training images, which keeps discriminator updates frequent enough over 25 epochs.

- lr = 0.0002: 2e-4 Adam for both networks; a lower rate than the VAE because adversarial training on 16,000 images is the least stable of the three families.

Training (cgan_s0): 25 of 25 max epochs, kept epoch 25 (planned epochs completed (GANs are not early-stopped)); 3.116 min on mps; 14400 fit / 1600 validation samples; 730,564 parameters

Training (cgan_s1): 25 of 25 max epochs, kept epoch 25 (planned epochs completed (GANs are not early-stopped)); 3.349 min on mps; 14400 fit / 1600 validation samples; 730,564 parameters

cdiffusion

Family: diffusion (conditional); input: transformed; seeds [0]; serves class_balancing; status planned; estimated 10 min/seed

Why this model: Conditional diffusion with classifier-free guidance usually gives the most realistic and class-faithful image samples, which is what a class-6 augmentation needs to help a tissue classifier. One seed because diffusion training is stable and the budget leaves no room for a second.

- timesteps = 400: 400 noise levels is enough for 28x28x3 patches and keeps per-epoch cost at ~12.5 s on ~14,400 images, so diffusion fits the remaining budget.

- base_channels = 32: 32 UNet channels; a wider UNet would roughly double the ~12.5 s/epoch cost and push diffusion past the 20-minute total.

- sampling_steps = 50: 50 DDIM steps (~0.02 s per image) so generating the several thousand class-6 images needed to balance against class 8 stays well under a minute.

- guidance_scale = 3: 3.0, toward the upper end of the 2-4 range: for balancing class 6 it matters most that samples are actually normal colon mucosa rather than a neighbouring class, at some cost in diversity. Judgment call; 2.0 would give more diversity but more label noise. Revision 2 note: observed class-6 agreement was 0.5755 vs 0.8974 on real

Models, hyperparameters, and training (cont.)

class-6 images, so 3.0 was not enough at 45 epochs; the proposed follow-up (cdiffusion_long) raises it to 4.0.

- epochs = 45: Maximum of 45 epochs (~9.4 min at ~12.5 s/epoch), the largest share left after the cVAE (2.5 min) and two GAN seeds (7 min) inside 20 minutes; diffusion keeps improving with training, so it gets the most time.

- batch_size = 128: 128 images per step, ~112 steps per epoch on ~14,400 training images, matching the other two models so differences are not due to batch size.

- lr = 0.0002: 2e-4 Adam, a conservative rate for the denoising UNet so the short 45-epoch run does not oscillate.

Training (cdiffusion_s0): 45 of 45 max epochs, kept epoch 44 (max epochs reached); 9.233 min on mps; 14400 fit / 1600 validation samples; 569,891 parameters

cdiffusion_long

Family: diffusion (conditional); input: transformed; seeds [0]; serves class_balancing; status not_run_budget; estimated 19 min/seed

Why this model: The follow-up the critique pointed to: diffusion was the only generator whose class-6 samples were mostly recognized as class 6, and it was still improving at its epoch cap. Not run because revision 1 already used about 17.4 of the 20 training minutes (cvae 1.745, cgan about 3.1 and 3.4, cdiffusion 9.233), which leaves no room for a longer diffusion run or a second diffusion seed.

- timesteps = 400: 400 noise levels as in cdiffusion, so only training length and guidance differ and any change in class-6 agreement can be attributed to them.

- base_channels = 32: 32 UNet channels as in cdiffusion (~12.5 s/epoch on ~14,400 PathMNIST images); widening would confound the comparison and double the cost.

- sampling_steps = 50: 50 DDIM steps as in cdiffusion, keeping generation of the 889 class-6 images needed for balancing well under a minute.

- guidance_scale = 4: 4.0, the top of the usual range, because cdiffusion_s0 at 3.0 produced recognizable class-6 images only 57.55% of the time (89.74% for real class-6 images); stronger guidance trades diversity for class fidelity, which is what balancing class 6 needs.

- epochs = 90: Maximum of 90 epochs (about 19 min at ~12.5 s/epoch), double cdiffusion's 45, because cdiffusion_s0's validation loss was still falling at the cap (best epoch 44 of 45); early stopping applies.

- batch_size = 128: 128 as in cdiffusion and the other arms, so the comparison isolates epochs and guidance.

- lr = 0.0002: 2e-4 Adam as in cdiffusion, whose loss curve decreased smoothly without oscillation over 45 epochs.

Evaluation focus

Models, hyperparameters, and training (cont.)

The decisive measurement is the downstream test on held-out real images: class-6 F1 (and macro-F1 as a guard) for real data, real + random oversampling of class 6, and real + each model's class-6 samples. Because the imbalance is only 1.634x, oversampling is the baseline to beat and a no-gain result is plausible. Sample fidelity (FID-style and C2ST metrics against held-out real images, with their real-vs-real references) and class fidelity of generated images explain why a model does or does not help; memorization distances are reported as evidence only.

Evaluation metrics (held-out real data)

Realism metrics use the 128-d feature space of the evaluation classifier (held-out accuracy 0.94); per-pixel statistics use all pixels. Each realism metric has a real-vs-real reference row.

Fidelity

	cvae_s0	cgan_s0	cgan_s1	cdiffusion_s0
n_compared	4000	4000	4000	4000
frechet_distance	264.6948	97.8199	71.1999	42.1907
frechet_distance_reference	0.2915	0.2915	0.1627	0.2915
mmd2	0.37598	0.11422	0.08677	0.03598
mmd2_reference	-0.0001	-0.0001	-0.00012	-0.0001
c2st_auc	0.9998	0.9997	0.9992	0.9934
c2st_auc_reference	0.5081	0.5081	0.502	0.5081
precision	0.674	0.5587	0.5465	0.627
recall	0.034	0.023	0.0437	0.2855

Conditional

	cvae_s0	cgan_s0	cgan_s1	cdiffusion_s0
label_agreement	0.1465	0.214	0.2407	0.8333
label_agreement_reference	0.94	0.94	0.94	0.94

Memorization

	cvae_s0	cgan_s0	cgan_s1	cdiffusion_s0
dcr_median_synthetic	5.0479	4.9148	4.3691	4.3575
dcr_median_heldout	2.4628	2.4628	2.4628	2.4628
dcr_ratio	2.0496	1.9956	1.774	1.7693
too_close_fraction	0	0.0003	0	0

Evidence about copying of training samples, not a formal privacy guarantee.

Memorization pixels

	cvae_s0	cgan_s0	cgan_s1	cdiffusion_s0
dcr_median_synthetic	1.4906	4.5557	5.1421	4.8338
dcr_median_heldout	4.1324	4.1324	4.1324	4.1324
dcr_ratio	0.3607	1.1024	1.2443	1.1697
too_close_fraction	0.038	0	0.0003	0

DCR in raw pixel space (scaled to [0, 1]): detects near-identical copies of training images.

Feature stats

	cvae_s0	cgan_s0	cgan_s1	cdiffusion_s0
mean_corr	0.9995	0.9222	0.9835	0.9936
var_corr	0.9849	0.9761	0.957	0.9779

Likelihood

	cvae_s0	cgan_s0	cgan_s1	cdiffusion_s0
train_mean	-1397.421	n/a	n/a	n/a

Evaluation metrics (held-out real data) (cont.)

heldout_mean	-1399.275	n/a	n/a	n/a
per_dimension_heldout	-0.5949	n/a	n/a	n/a

Likelihoods are in each model's own input space and units; do not compare them across models with different input spaces.

Objective test: class balancing on held-out real data

Target class(es): 6. Each was topped up to the largest training class (6: +889 images). Classifier: SmallCNN trained for exactly 20 epochs (cosine learning-rate decay, final weights) on each training set.

training set	n_train	target F1	macro F1	accuracy
real_seed0	16000	0.9175	0.9337	0.9337
real_seed1	16000	0.9112	0.9358	0.9355
real_plus_resampled	16889	0.9191	0.9366	0.937
real_plus_cvae_s0	16889	0.9028	0.933	0.9335
real_plus_cgan_s0	16889	0.9125	0.9359	0.9363
real_plus_cgan_s1	16889	0.9107	0.932	0.9325
real_plus_cdiffusion_s0	16889	0.9072	0.9344	0.9347

Noise level (target-F1 difference between the two real-only seeds): 0.0063. noise_level_target_f1 is the target-class F1 difference between two real-only runs that differ only in the classifier's seed. A change smaller than this is indistinguishable from training noise. Two runs are a rough yardstick, not a confidence interval.

Definitions

- target_f1: F1 score (harmonic mean of precision and recall) for the target class(es), on held-out real data.
- macro_f1: Unweighted mean of per-class F1 on held-out real data; checks that helping the target class did not hurt others.
- noise_level_target_f1: Target-class F1 difference between two real-only classifier runs with different seeds.

How to read the metrics

- mmd2: Squared maximum mean discrepancy (RBF kernel, bandwidth = median distance among held-out real samples) between synthetic and held-out real samples in the evaluation PCA space. 0 = same distribution; compare with mmd2_reference. The estimator is unbiased, so values near 0 can be slightly negative.

- mmd2_reference: Same MMD, but a training-set subsample vs held-out real: the value two real samples of this size give.

- c2st_auc: Classifier two-sample test: cross-validated ROC AUC of a gradient-boosted classifier separating synthetic from held-out real samples. 0.5 = indistinguishable, 1.0 = perfectly separable.

- c2st_auc_reference: Same test, training subsample vs held-out real (should be near 0.5).

- precision: Fraction of synthetic samples inside the 5-NN radius of some held-out real sample (fidelity: are samples realistic?).

- recall: Fraction of held-out real samples inside the 5-NN radius of some synthetic sample (coverage: is the data's variety reproduced?).

- dcr_median_synthetic: DCR = distance to closest record: for one sample, the Euclidean distance (evaluation space) to its nearest training sample. This is the median DCR over synthetic samples.

- dcr_median_heldout: Median DCR over held-out real samples: how close a genuinely new sample from the same population typically is to some training sample. This is the yardstick for the synthetic DCR.

- dcr_ratio: dcr_median_synthetic / dcr_median_heldout. About 1 means synthetic samples are as far from the training data as new real samples are; well below 1 means they sit unusually close to training samples (possible copying); above 1 means they are farther away than real data (often a sign of off-distribution samples, not of privacy).

- too_close_fraction: Fraction of synthetic samples whose DCR is below the 5th percentile of held-out DCRs. If the model does not copy, about 0.05 is expected (the same share as for new real samples); much higher means a group of synthetic samples are near-copies of training samples. Evidence about copying only, not a privacy guarantee.

- mean_corr: Pearson correlation, across features, between per-feature means of synthetic and held-out real data (transformed space).

- var_corr: Same, for per-feature variances.

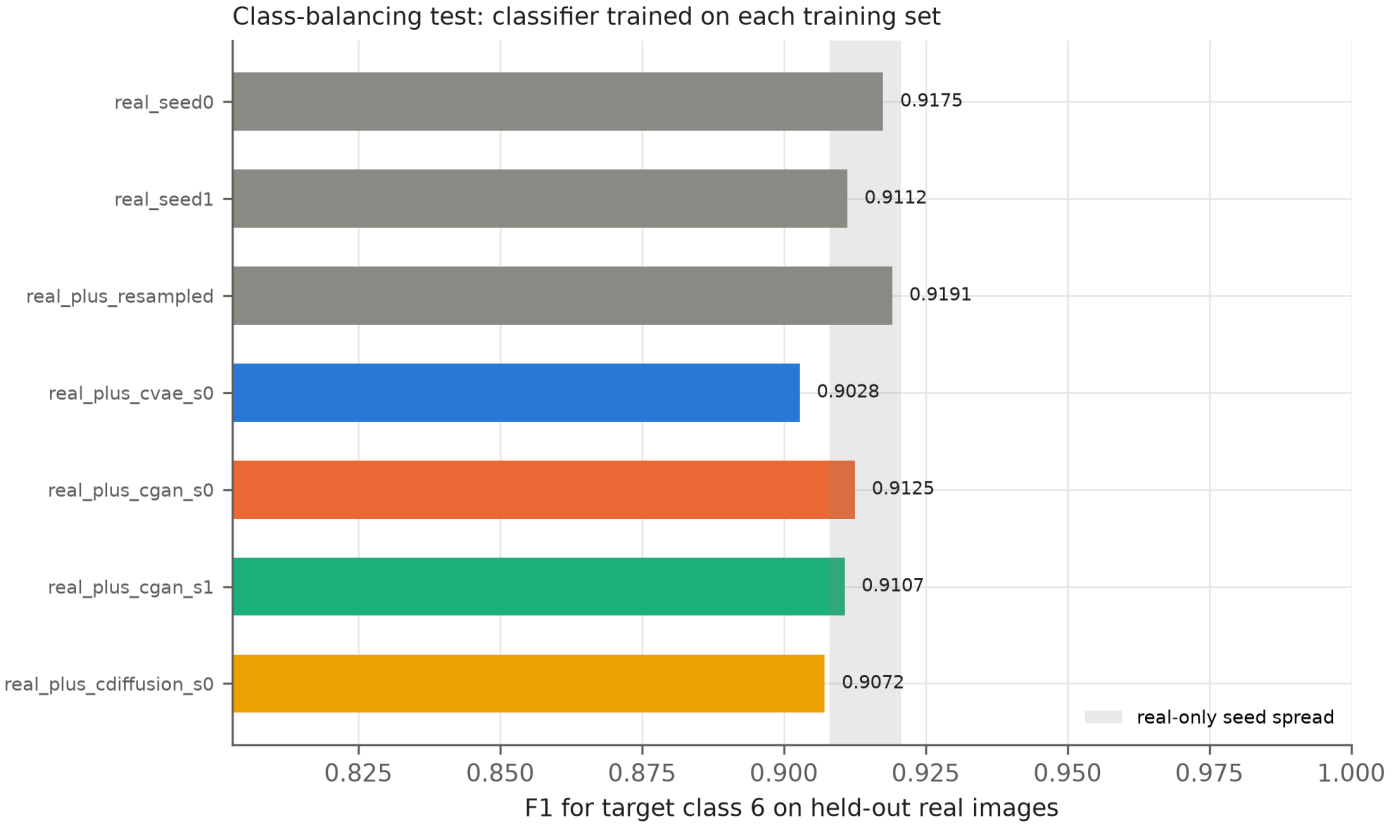
- train_mean: Mean per-sample ELBO (VAE, a lower bound) or exact log-likelihood (flow) on training data, in the model's own input space. Not comparable across models with different input spaces.

- heldout_mean: Same quantity on held-out real data; a large drop from train_mean indicates overfitting.

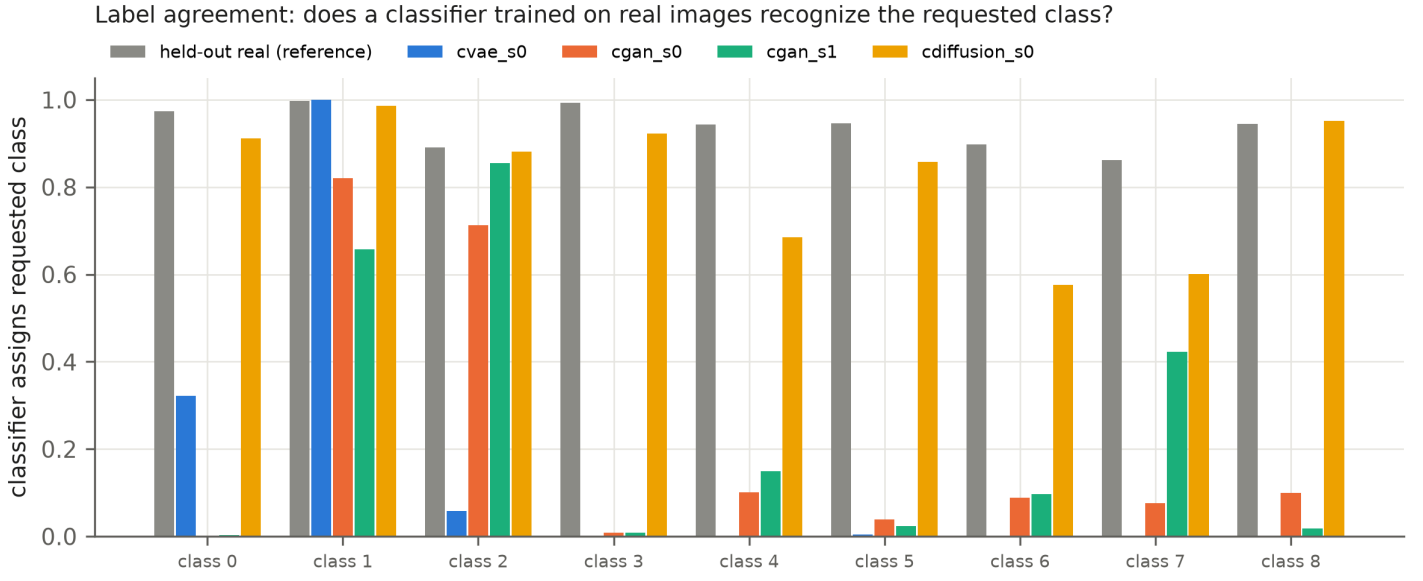
- frechet_distance: Frechet distance between Gaussians fit to synthetic and held-out real features (the FID formula, computed on the evaluation classifier's features instead of Inception's). 0 = same mean and covariance; compare with frechet_distance_reference.

How to read the metrics (cont.)

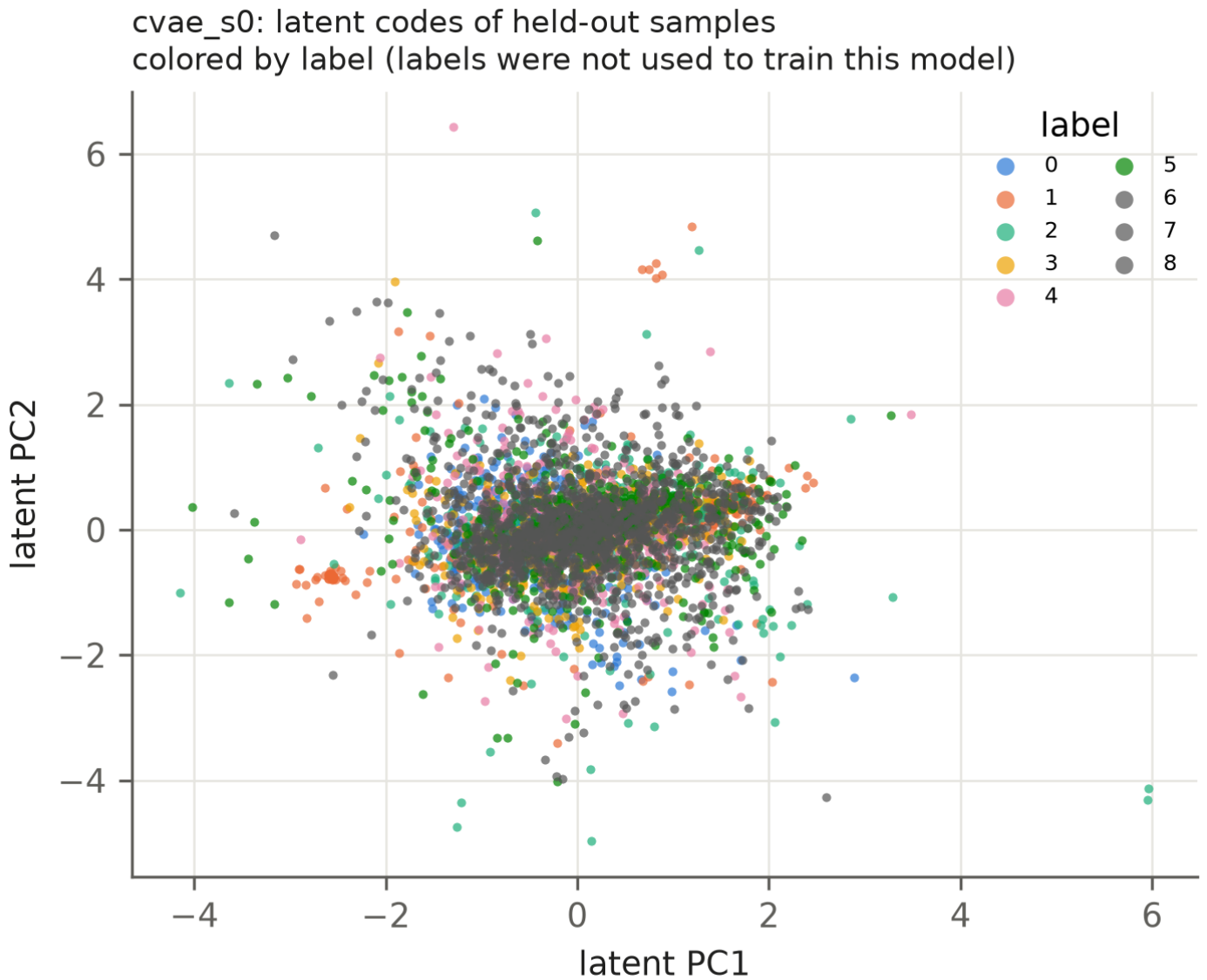
- frechet_distance_reference: Same distance, training-set subsample vs held-out real: what two real samples of this size give.
- label_agreement: Share of synthetic images that the evaluation classifier (trained on real images) assigns to the class they were generated for.
- label_agreement_reference: The classifier's accuracy on held-out real images: the agreement a perfect class-conditional generator would reach.



Class-balancing test: F1 for the target class on held-out real images, for a classifier trained on each training set.
The gray band spans the two real-only runs (training noise).

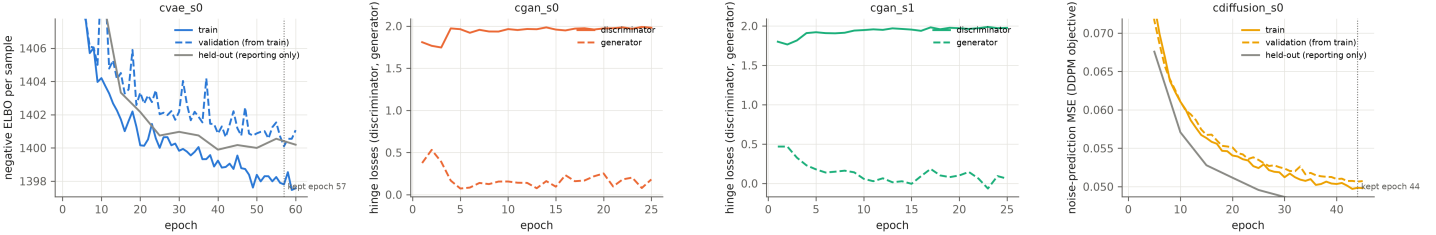


Per class: share of each model's images that the evaluation classifier assigns to the requested class; gray bars are the classifier's accuracy on held-out real images (the ceiling).

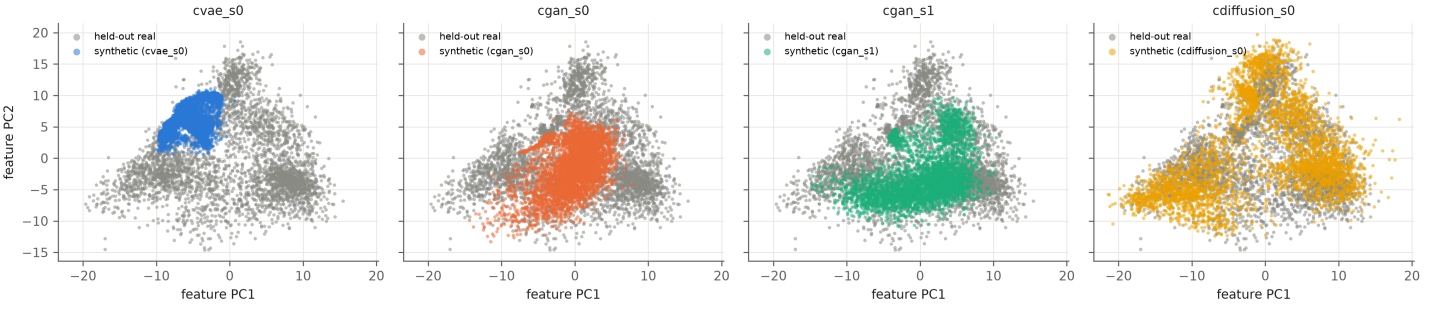


Two-dimensional PCA of a VAE's latent codes for held-out samples. A picture for orientation; neighbor-preservation numbers are on the metrics page.

Training curves

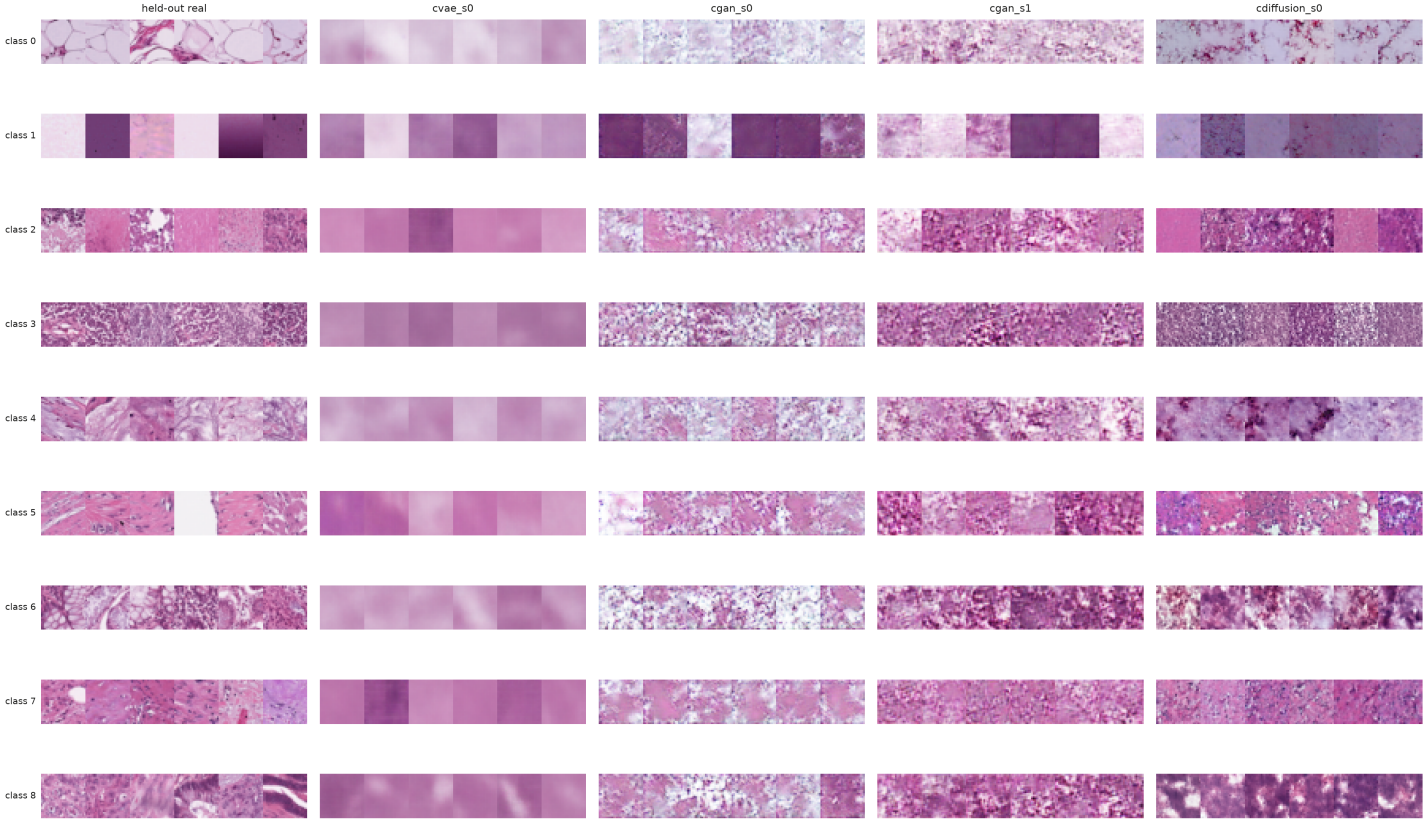


Training curves. The validation set is 10% of the training split and decides early stopping; the held-out curve is shown for reference only and never used to choose anything.



Held-out real samples (gray) and synthetic samples in the first two principal components of the shared evaluation space (for images: of the evaluation classifier's features). Overlap in two components is necessary but not sufficient; the C2ST and MMD metrics use all components.

Held-out real images and generated images, by class (rows)



Held-out real images (first column block) and each model's samples, one row per class. Conditional models were asked for the class of the row; unconditional models' samples are not sorted by class.

Findings

What each model produced

cdiffusion_s0 (conditional diffusion, 45 epochs, 9.233 min). This is the only generator whose samples mostly have the requested class. Overall label agreement was 0.8333 (reference 0.94, the evaluation classifier's accuracy on held-out real images). For class 6 it was 0.5755, against 0.8974 on real class-6 images. Fidelity was still far from real data:

- Fréchet distance 42.1907 (real-vs-real reference 0.2915).
- C2ST AUC 0.9934 (reference 0.5081).
- MMD² 0.03598 (reference -0.0001).
- Precision 0.627, recall 0.2855.

Training hit the epoch cap with validation loss still falling (best epoch 44). The pixel-space DCR ratio was 1.1697 with a too-close fraction of 0.0, so there is no sign of near-copies.

cgan_s0 and cgan_s1 (conditional GAN, 25 epochs each). Both seeds failed to condition on the class:

- Label agreement 0.214 and 0.2407 (reference 0.94).
- Class-6 agreement 0.0883 and 0.0969.
- Fréchet distance 97.8199 (reference 0.2915) and 71.1999 (reference 0.1627).
- C2ST AUC 0.9997 and 0.9992 (references 0.5081 and 0.502).
- Recall 0.023 and 0.0437.

In both seeds the discriminator's hinge loss rose toward 2, which means it had stopped separating real from fake. The conditioning failure holds for both seeds. Pixel DCR ratios were 1.1024 and 1.2443.

cvae_s0 (conditional VAE, 60 epochs, best epoch 57). Its samples are blur that ignores the label:

- Label agreement 0.1465 (reference 0.94). Class-6 agreement is 0.0, and every class-1 (background) sample is labeled correctly.
- Fréchet distance 264.6948 (reference 0.2915).
- C2ST AUC 0.9998 (reference 0.5081).

The held-out ELBO (-1399.275) is close to the training value (-1397.421), so the model is not overfit. Its pixel-space DCR ratio is 0.3607 (too-close fraction 0.038). This reflects smooth, low-variance samples sitting near smooth training images, not copies: in the classifier feature space its DCR ratio is 2.0496.

cdiffusion_long (90 epochs, guidance 4.0) was added in revision 2 as the follow-up the critique suggested. It was not run because of the budget.

What the figures show

- Sample grid: real patches show clear tissue structure, for example fat-cell outlines in class 0 and glands in class 6. The cVAE rows are flat pink blur in every class. The two GAN columns show similar mottled pink-white texture regardless of the requested class. Diffusion samples have the most structure and differ by class (fat-like class 0, dense dotted lymphocytes in class 3), but its class-6 images do not show the gland structure of

Findings (cont.)

real mucosa.
- Real-vs-synthetic PCA plot: diffusion samples spread over all three lobes of the real data. cvae_s0 covers one small patch. The GAN samples sit in the middle and miss the edges.
- Label-agreement chart: this shows the per-class pattern above. Diffusion is close to the real reference for most classes but low for classes 4, 6 and 7.
- Downstream chart: every synthetic bar sits at or left of the real-only seed spread, and oversampling is the highest bar.
- Loss curves: the diffusion validation loss was still falling at the end of training. The cVAE validation loss had flattened. The GAN discriminator losses sit near 2.
- cVAE latent plot: classes overlap, apart from a small class-1 cluster.

How the models compare

All four models were scored in the same space: the 128-d features of one evaluation classifier trained on real images.

- Fidelity and coverage: cdiffusion_s0 is best on every shared metric (Fréchet distance, MMD^2, C2ST, recall, label agreement). cvae_s0 is worst.
- C2ST is saturated: it is at least 0.9934 for every model, so it cannot rank them.
- Downstream test: this ranking does not carry over. The class-6 F1 spread among synthetic arms is close to the classifier noise level of 0.0063, and each arm was trained with one classifier seed, so the generators cannot be ranked on that test.
- Likelihoods: only the cVAE reports one (an ELBO), so likelihoods are not compared.

Answer to the objective

No. The success criterion was a class-6 F1 above real data plus random oversampling, without lowering macro-F1. Every synthetic arm fell short:

- Oversampling: class-6 F1 0.9191, macro-F1 0.9366.
- real_plus_cdiffusion_s0: class-6 F1 0.9072, macro-F1 0.9344.
- real_plus_cgan_s0: class-6 F1 0.9125, macro-F1 0.9359.
- real_plus_cgan_s1: class-6 F1 0.9107, macro-F1 0.932.
- real_plus_cvae_s0: class-6 F1 0.9028, macro-F1 0.933.

For the cVAE and both GAN seeds, this result reflects mostly mislabeled added images rather than a fair test of synthesis. Only cdiffusion_s0 actually tested the idea, and it also fell short. Oversampling's own gain over real data alone (0.9175) is within noise. A 1.634x imbalance with 1,402 real class-6 training images appears not to limit this classifier much.

Downstream use of the generated data

Add conditional-diffusion class-6 images to the real training set to raise class-6 F1

Verdict: NOT_RECOMMENDED | task: class_balancing | model: cdiffusion_s0

This is the only generator whose class-6 samples were mostly recognized as class 6, yet adding them gave a lower class-6 F1 than random oversampling of real class-6 images, by more than the classifier's seed-to-seed noise. A classifier also separates its samples from held-out real images almost perfectly.

- evidence: downstream.results.real_plus_cdiffusion_s0.target_f1 = 0.9072 (vs downstream.results.real_plus_resampled.target_f1 = 0.9191; downstream.noise_level_target_f1 = 0.0063)

- evidence: downstream.results.real_plus_cdiffusion_s0.macro_f1 = 0.9344 (vs downstream.results.real_plus_resampled.macro_f1 = 0.9366)

- evidence: cdiffusion_s0.conditional.label_agreement = 0.8333 (vs cdiffusion_s0.conditional.label_agreement_reference = 0.94; class-6 per_class_agreement 0.5755 vs per_class_heldout_accuracy 0.8974)

- evidence: cdiffusion_s0.fidelity.c2st_auc = 0.9934 (vs cdiffusion_s0.fidelity.c2st_auc_reference = 0.5081)

- caveat: One generator seed and one classifier seed per arm; the noise level comes from only two real-only runs.

- caveat: Training stopped at the 45-epoch cap with validation loss still falling; a longer, higher-guidance run (cdiffusion_long, not run for budget) might change the result.

- caveat: About 42% of the added 'class 6' images are not recognized as class 6 by the evaluation classifier, so the arm mixes real augmentation with label noise.

Add conditional-GAN class-6 images to the real training set to raise class-6 F1

Verdict: NOT_RECOMMENDED | task: class_balancing | model: cgan_s0

Both GAN seeds failed to produce the requested class (class-6 agreement 0.0883 and 0.0969), so the added images are mostly mislabeled. Both seeds scored below oversampling on class-6 F1, so the conclusion holds for both seeds.

- evidence: downstream.results.real_plus_cgan_s0.target_f1 = 0.9125 (vs downstream.results.real_plus_resampled.target_f1 = 0.9191; downstream.noise_level_target_f1 = 0.0063)

- evidence: downstream.results.real_plus_cgan_s1.target_f1 = 0.9107 (vs downstream.results.real_plus_resampled.target_f1 = 0.9191)

- evidence: cgan_s0.conditional.label_agreement = 0.214 (vs cgan_s0.conditional.label_agreement_reference = 0.94)

- evidence: cgan_s1.conditional.label_agreement = 0.2407 (vs cgan_s1.conditional.label_agreement_reference = 0.94)

- caveat: The discriminator loss rose toward its no-signal value in both seeds; different

Downstream use of the generated data (cont.)

GAN settings might condition properly, but that was not tested.

Add conditional-VAE class-6 images to the real training set to raise class-6 F1

Verdict: NOT_RECOMMENDED | task: class_balancing | model: cvae_s0

The cVAE produces featureless blur that ignores the class label (class-6 agreement 0.0), and it gave the lowest class-6 F1 of all arms, below oversampling by more than the noise level.

- evidence: downstream.results.real_plus_cvae_s0.target_f1 = 0.9028 (vs downstream.results.real_plus_resampled.target_f1 = 0.9191; downstream.noise_level_target_f1 = 0.0063)

- evidence: cvae_s0.conditional.label_agreement = 0.1465 (vs cvae_s0.conditional.label_agreement_reference = 0.94)

- evidence: cvae_s0.fidelity.frechet_distance = 264.6948 (vs cvae_s0.fidelity.frechet_distance_reference = 0.2915)

- caveat: Kept only as a negative control.

Balance class 6 by random oversampling of real class-6 training images instead of synthes

Verdict: CONDITIONAL | task: class_balancing | model: cdiffusion_s0

Oversampling scored highest of every arm on class-6 F1 and did not lower macro-F1, so it is the better choice than any generator here. Its gain over real data alone is within the classifier's noise, so with a 1.634x imbalance, balancing may not be needed at all.

- evidence: downstream.results.real_plus_resampled.target_f1 = 0.9191 (vs downstream.results.real_seed0.target_f1 = 0.9175; downstream.results.real_seed1.target_f1 = 0.9112; downstream.noise_level_target_f1 = 0.0063)

- evidence: downstream.results.real_plus_resampled.macro_f1 = 0.9366 (vs downstream.results.real_seed0.macro_f1 = 0.9337)

- caveat: This baseline does not depend on any generator; model_key is filled in only because the schema requires one.

- caveat: Use it only as the cheap default; it has not been shown to improve on training on real data alone.

Not tested (no recommendation either way)

- Whether a longer, higher-guidance diffusion run (cdiffusion_long, planned but not run for budget) would beat oversampling; that would need about 19 more training minutes plus the downstream test.

- Whether adding only the synthetic class-6 images that the evaluation classifier recognizes as class 6 helps; the pipeline has no filtered downstream arm.

- Stable ranking of the arms: each was trained with one classifier seed, and at least 3 seeds per arm would be needed.

- Performance on the CRC-VAL-HE-7K test images from a different clinic, which this project

Downstream use of the generated data (cont.)

does not load; all scores are on held-out images from the same NCT-CRC-HE-100K source.

- Balancing all 9 classes, or any class other than class 6.

Privacy

The objective does not involve sharing or releasing data. There is no formal privacy guarantee (such as differential privacy) for any synthetic image here. The memorization checks (median distance to the closest training image, pixel-space dcr_ratio 0.3607 for cvae_s0 and between 1.1024 and 1.2443 for the other models) are evidence about near-copying only.

Critique and revision

An independent critic subagent reviewed the objective, plan, metrics, and training logs (not the agent's own reasoning) before the recommendation and findings were written.

The downstream test measures the success criterion directly, and it fails for every model: all four synthetic arms score below random oversampling on class-6 F1 (0.9028-0.9125 vs 0.9191). That much is supported. Three things weaken what else might be concluded. First, three of the four generators did not produce class-6 images at all (class-6 label agreement 0.0 for cVAE, 0.088 and 0.097 for the two cGAN seeds). For those arms the test measures label noise, not synthetic balancing, so it says little about whether good synthetic class-6 images would help. Second, the gaps between synthetic arms (0.0097 range on class-6 F1) are about the size of the classifier noise (0.0063), so ranking models or naming a 'best' one is not supported. Third, diffusion is the only usable generator, it rests on one seed, and its validation loss was still falling at the epoch cap.

[high] objective test

The downstream test does compare against both real-only and oversampling baselines, as the criterion requires. But for cVAE and both cGAN seeds, almost none of the 889 added 'class 6' images look like class 6 to the evaluation classifier. The result therefore cannot support 'synthetic class-6 images do not beat oversampling' as a statement about synthesis in general. Only the diffusion arm actually tests the objective, and even there 42% of the added images are not recognized as class 6.

Evidence: Class-6 per_class_agreement: cvae_s0 0.0, cgan_s0 0.0883, cgan_s1 0.0969, cdiffusion_s0 0.5755. The reference is per_class_heldout_accuracy['6'] = 0.8974 (metrics/*.json). Downstream target_f1: real_plus_resampled 0.9191, cvae 0.9028, cgan_s0 0.9125, cgan_s1 0.9107, cdiffusion 0.9072 (metrics/downstream.json). The sample grid (figures/sample_grid.png) shows the cVAE class-6 row as featureless pink blur and both cGAN class-6 rows as class-agnostic texture with no glandular structure.

Recommendation: Report the conclusion as 'no gain from these generators'. Only the diffusion arm counts as a real test of synthetic balancing. Optionally, add a downstream arm that keeps only the diffusion class-6 samples the evaluation classifier labels as class 6 (a filtered arm), so the test separates generator quality from label noise.

[high] seed stability

The differences in class-6 F1 between synthetic arms are about the same as the classifier's own seed-to-seed noise. Each arm, and the oversampling baseline, was trained with a single classifier seed. Ranking the generators by downstream F1 (for example, cGAN 'best' downstream) is therefore unsupported. The downstream order also reverses the fidelity order (diffusion is best on every fidelity metric), which is what noise would produce.

Evidence: noise_level_target_f1 = 0.0063, from only two real-only runs (0.9175 vs 0.9112). cgan_s0 0.9125 vs cdiffusion_s0 0.9072 is a gap of 0.0053, below the noise. The full synthetic range is 0.9028-0.9125 (0.0097, about 1.5x the noise). Oversampling's gain over real-only (0.9191 vs 0.9175) is 0.0016, also inside the noise. Macro-F1 across all arms spans 0.9320-0.9366. Two real seeds differ by 0.0021 (metrics/downstream.json).

Recommendation: Run each downstream arm with at least 3 classifier seeds, including real_plus_resampled, and report the mean and spread. Classifier training is cheap compared

Critique and revision (cont.)

with generator training. Until then, state only that no synthetic arm beat oversampling. Do not rank the generators.

[medium] seed stability

The two cGAN seeds differ substantially on fidelity and on per-class conditioning. Diffusion and cVAE have one seed each, so any statement about diffusion (the only usable generator) rests on a single run.

Evidence: cgan_s0 vs cgan_s1: frechet_distance 97.82 vs 71.20, mmd2 0.114 vs 0.087, label_agreement 0.214 vs 0.241, class-7 agreement 0.077 vs 0.423, class-8 agreement 0.0995 vs 0.0175 (metrics/cgan_s*.json). The cGAN seed gap in FD (26.6) is about half the cGAN-to-diffusion gap (29-56). The fidelity ranking of diffusion over cGAN holds across both cGAN seeds. The single cdiffusion_s0 run is the only one with usable class-6 samples.

Recommendation: Move budget from the cVAE and the cGAN to a second diffusion seed (see the training finding). Treat cGAN per-class results as unstable.

[high] training adequacy

Both cGAN runs show a discriminator that has stopped discriminating: the hinge D loss climbs toward 2.0, the value it takes when D outputs about 0 on real and fake alike. Class conditioning failed: overall label agreement is near chance, and class 0 is never produced. More epochs at these settings are unlikely to fix this.

Evidence: runs/cgan_s0.json: d_loss rises from 1.75 (epoch 3) to 1.98-1.99 (epochs 22-25), and g_loss stays at 0.08-0.25. runs/cgan_s1.json: d_loss is 1.975 at epoch 25, and g_loss goes negative (-0.0045 at epoch 15, -0.0646 at epoch 23). Label agreement is 0.214 and 0.241 against a 0.94 reference. Class-0 agreement is 0.0 and 0.0024. C2ST AUC is 0.9997 and 0.9992.

Recommendation: Either drop the cGAN and give its ~6.5 minutes to diffusion, or change the GAN settings (for example, a higher discriminator learning rate or more D steps per G step) and keep its result labelled exploratory.

[medium] training adequacy

The diffusion model, the only generator with usable class-6 samples, hit its epoch cap with validation loss still falling. Its weakest classes include class 6, the target. More training might improve class-6 fidelity and change the downstream result for the one arm that matters.

Evidence: runs/cdiffusion_s0.json: stop_reason 'max epochs reached', best_epoch 44 of 45. val_loss went from 0.05179 (epoch 35) to 0.05066 (epoch 44), and heldout_loss from 0.04808 (epoch 35) to 0.04696 (epoch 45). The run used 9.23 min. Total training was about 17.4 of 20 min (cvae 1.75, cgan 3.12 + 3.35, diffusion 9.23).

Recommendation: Within the 20-minute budget, drop the cVAE (unusable samples) and the cGAN (failed conditioning). Use the ~8 freed minutes for diffusion: either about 80 epochs on one seed, or two seeds of about 45 epochs each.

[medium] training adequacy

Critique and revision (cont.)

The cVAE reached its epoch cap, but its validation loss had plateaued, so more epochs would not help. The real failure is that its samples carry no class information: the KL term is about 4.7 nats across 32 latent dimensions, and the samples are uniform blur that the classifier reads mostly as class 1 (background).

Evidence: runs/cvae_s0.json: epochs_run 60, best_epoch 57. val_loss went from 1400.87 (epoch 40) to 1400.10 (epoch 57), and kl is 4.78 at epoch 60. metrics/cvae_s0.json: per_class_agreement is 1.0 for class 1 and 0.0 for classes 3, 4, 6, 7 and 8. C2ST AUC 0.9998, recall 0.034.

Recommendation: Drop the cVAE from the class-balancing plan, or keep it only as a stated negative control. Do not count it as evidence about synthetic balancing.

[low] references

The references are sound: C2ST reference near 0.5, Fréchet and MMD references tiny compared with the model values, and n_compared = 4000. However, C2ST is saturated (≥0.993) for every model, so it cannot rank them; the ranking rests on FD, MMD and recall. cgan_s1 also reports different reference values from the other three models, which suggests a different reference subsample.

Evidence: c2st_auc_reference 0.5081 (cvae, cgan_s0, cdiffusion) vs 0.502 (cgan_s1). frechet_distance_reference 0.2915 vs 0.1627. Model FD values range from 42.19 to 264.69. c2st_auc values: 0.9998, 0.9997, 0.9992, 0.9934 (metrics/*.json).

Recommendation: Use one fixed reference subsample for all models. Do not cite C2ST as a basis for ranking models.

[low] memorization

The cVAE's pixel-space DCR ratio is well below 1, which on its face is a copying flag. The too_close_fraction is under 0.05, though, and the feature-space ratio is about 2. The low pixel ratio most likely means blurry, low-variance samples sit near smooth training images such as background, not copies. The objective does not mention sharing or privacy, so no privacy claim is at stake.

Evidence: metrics/cvae_s0.json memorization_pixels: dcr_ratio 0.3607, too_close_fraction 0.038. memorization: dcr_ratio 2.0496, too_close_fraction 0.0. All other models have pixel dcr_ratio between 1.10 and 1.24 and too_close_fraction ≤ 0.0003.

Recommendation: no plan-level fix: state as a limitation. Note that the cVAE's low pixel DCR comes from blur, not memorization.

[medium] unjustified choices

Several reasons are contradicted by the metrics. The cGAN reason expects sharper class-6 patches. The diffusion guidance_scale reason expects samples that are 'actually normal colon mucosa', yet class 6 is among diffusion's worst classes. The cGAN two-seed reason anticipates a collapsed run, but both seeds failed conditioning. Other reasons are generic, and the cVAE likelihood reason would read the same on any [0,1] image dataset.

Evidence: plan.json cgan reason: 'Conditional GAN can produce sharper class-6 patches than

Critique and revision (cont.)

the VAE, which may matter for a classifier distinguishing normal mucosa from adenocarcinoma epithelium'. Observed class-6 agreement is 0.088 and 0.097. cdiffusion guidance_scale reason: 'it matters most that samples are actually normal colon mucosa rather than a neighbouring class'. Observed class-6 agreement is 0.5755 (vs 0.8974 reference); only classes 4 (0.6854) and 7 (0.6005) are comparably low. cvae likelihood reason: 'pixel intensities are bounded, which a Gaussian decoder would not respect' is generic and ignores that H&E pixels are continuous, not binary. cvae lr reason: 'the cVAE is the most stable family here' is generic.

Recommendation: In the revised plan, justify guidance_scale using the observed class-6 agreement (for example, test a higher guidance or filter samples by classifier agreement). Drop or reframe the cGAN and cVAE reasons so they match their observed failure to condition.

Revision made in response

Plan revision 2: Revision 2 addresses critique.json without retraining, because revision 1 had already used about 17.4 of the 20-minute training budget. (1) The cVAE and cGAN reasons now record that they failed to condition (class-6 label agreement 0.0, 0.0883, 0.0969), so their downstream arms are reported as negative controls and label-noise tests, not as evidence about synthetic balancing. (2) The longer, higher-guidance diffusion run the critique recommended is added as cdiffusion_long with status not_run_budget and its time estimate. (3) The guidance_scale and cVAE likelihood/lr reasons were revised to match the observed results. Findings with no plan-level fix in this pipeline (more downstream classifier seeds, a classifier-filtered synthetic arm, a shared reference subsample) are stated as limitations in findings.md.

Limitations

- cdiffusion_long_s0 was planned but not run because it did not fit the time budget.
- Image realism is judged in the feature space of a classifier trained on the same real data; differences that do not affect classification (fine texture, staining details) are not measured by these numbers.
- The class-balancing test uses one classifier architecture and one seed per augmented training set; differences below the real-only seed spread (0.0063) are not meaningful.
- One plan revision, no retraining: the critique's main fixes (more diffusion training, a second diffusion seed) did not fit the remaining 20-minute training budget. They were recorded as cdiffusion_long with status not_run_budget.
- Fixes this pipeline does not support, stated here instead: training each downstream arm with 3 or more classifier seeds; adding only the synthetic class-6 images the classifier recognizes as class 6; and using one shared reference subsample for all models (cgan_s1's references differ from the others).
- Weak noise estimate: the noise level comes from only two real-only runs. It is a rough yardstick, not a confidence interval.
- Same-source evaluation: all evaluation is on held-out images from the same source as training. The different-clinic test set was not used, so stain shift between centers was not tested.
- Resolution: images are 28x28 downsamples, which lose fine histological texture.
- Budget: the 20-minute budget counts training only. The first full run's wall-clock time, including evaluation and the downstream test, was longer than that.
- No privacy assessment: memorization numbers are evidence about near-copying only. No privacy claim is made.

Reproducibility

Command: ./run.sh pathmnist "Balance the rarest tissue class to improve a classifier" --report-number 2

Plan seed 0; plan revision 2; budget 20 min; pipeline time 7.1 min

Preprocessing hash fa3206903097716f

python 3.13.5, torch 2.14.0, numpy 2.5.3, scikit-learn 1.9.1

- cvae_s0: seed 0, config hash 8dc688a92041778f, 1.745 min on mps

- cgan_s0: seed 0, config hash a5b8bc1ccff2123c, 3.116 min on mps

- cgan_s1: seed 1, config hash cf2bf05ab37136a1, 3.349 min on mps

- cdiffusion_s0: seed 0, config hash fabb75e60d9d5ab8, 9.233 min on mps

Generated 2026-10-25 13:13. All intermediate files: outputs/pathmnist/.